

BENCHMARK: ACCURATE — fit consistent with published values

Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456615.89738 +/- 0.00026 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1334 +/- 0.0076                 |
| Transit depth (Rp/Rs)^2                           | 1.78 +/- 0.2 [%]                  |
| Orbital Inclination (inc)                         | 88.04 +/- 0.86                    |
| Airmass coefficient 1 (a1)                        | 1.2123 +/- 0.0082                 |
| Airmass coefficient 2 (a2)                        | -0.1212 +/- 0.006                 |
| Scatter in the residuals of the lightcurve fit is | 0.67 %                            |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 3.77                              |
| Optimal Annulus                                   | 10.14                             |
| Transit Duration (day)                            | 0.1321 +/- 0.004                  |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                             |
|---------------------------|---------------------------------------------|
| Rp/R■ fit vs published    | 0.1334 ± 0.0076 vs 0.1241 (deviation 1.22σ) |
| Duration fit vs published | 0.1321 d vs 0.1317 d (deviation 0.11σ)      |
| Mid-time O–C              | -0.5 min                                    |
| Depth SNR                 | 17.6                                        |
| Residual scatter          | 0.67 %                                      |

Light curve

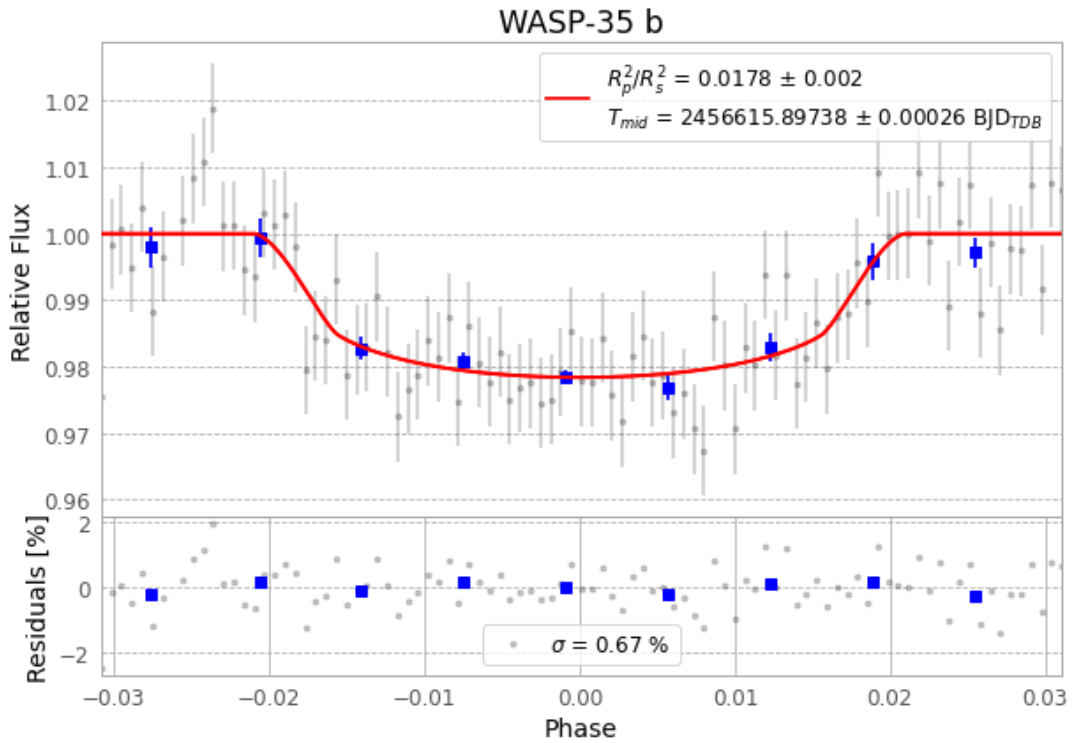


Figure 1 — FinalLightCurve\_WASP-35 b\_2013-11-19.png (SHA-256 0f977da6a506b1a5...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [264, 230], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 ef5ad4f2d1fd91b6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

95 FITS frames (62 MB), WASP-35131119070313.FITS → WASP-35131119114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-131119.log (5f7196ea1250...), FinalLightCurve\_WASP-35 b\_2013-11-19.png (0f977da6a506...), FinalLightCurve\_WASP-35 b\_2013-11-19.csv (0fe9c0a2dad9...)

## Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 23:37Z.